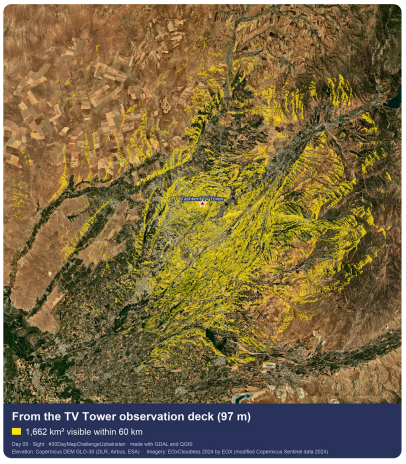
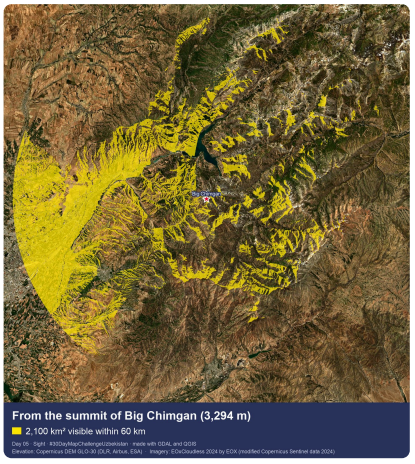
👁️ 05-kun · Ko‘rish: Toshkent teleminorasidan nimalarni ko‘rish mumkin?

GDAL bo‘yicha yangi boshlovchilar uchun dars. GDAL deyarli har bir xarita dasturi ichida ishlaydigan bepul buyruqlar satri vositalari to‘plami. Terminaldagi ilk qadamlaringiz: kod yozish tajribasi kerak emas.

Siz bepul **sun’iy yo‘ldosh balandlik ma’lumotlarini** (Copernicus DEM) yuklab olasiz, tayllarni bir-biriga ulaysiz, ulardan soyali relyef yasaysiz va kompyuterga bitta savol berasiz: **"Bu yerdan nimani ko‘ra olaman?"** Javob **ko‘rinish zonasi (viewshed)** deb ataladi. Toshkent teleminorasining etagidan va kuzatuv maydonchasidan ochiladigan manzarani solishtirasiz, **Katta Chimyon** cho‘qqisidan atrofga qaraysiz va ikkalasi bir-birini ko‘ra oladimi, tekshirasiz. Oxirida hammasini QGIS’da **Sentinel-2 sun’iy yo‘ldosh tasviri** ustiga joylashtirasiz.

⏱️ Taxminan 45 daqiqa (va GDAL’ni o‘rnatish uchun 10 daqiqa) · 💻 Windows, macOS yoki Linux · 🔑 Akkaunt kerak emas

📄 **Chop etish uchun PDF:** English · Русский · O‘zbekcha

Yerdan (1.7 m)	Maydonchadan (97 m)	Katta Chimyondan
		
264 km² ko‘rinadi	**1,662 km²** ko‘rinadi	**2,100 km²** ko‘rinadi

📚 0-qism · Asosiy tushunchalar

Balandlik modeli (DEM). Har bir pikselida metrdagi balandlik saqlanadigan raster (piksellar to‘ri). Bizniki **Copernicus DEM GLO-30**: har 30 m da bitta piksel. U 2011–2015-yillarda yonma-yon uchgan ikkita nemis radar sun’iy yo‘ldoshi, **TerraSAR-X** va **TanDEM-X** yordamida koinotdan o‘lchangan. Bu *sirt* modeli: unga binolar va daraxtlar ham kiradi, aynan ular ko‘rinishni to‘sadi.

Ko‘rinish zonasi. Biror joyda turib, har tomonga qarang: ko‘rgan hamma narsangiz ko‘rinish zonangiz bo‘ladi. Kompyuter ko‘zingizdan har bir pikselgacha ko‘rinmas to‘g‘ri chiziq tortadi. Agar yo‘lda tepalik, bino yoki o‘rmon bo‘lsa, u piksel yashirin hisoblanadi. Ko‘rinadigan piksellar **255** qiymatini, yashirinlari **0** qiymatini oladi.

Kuzatuvchi balandligi. Ko‘zingiz yerdan taxminan **1.7 m** balandda. Teleminoraning kuzatuv maydonchasida esa **97 m** balandda. Bir necha metr hammasini o‘zgartirib yuborishi mumkin.

Terminal. Tugmalarni bosish o‘rniga buyruqlar yoziladigan oyna. Har bir GDAL buyrug‘i bitta qatordan iborat: **vosita nomi**, `–` bilan boshlanadigan bir nechta **parametrlar**, so‘ng **kirish** va **chiqish** fayllari. Masalan: `gdaldem hillshade dem_utm.tif hillshade.tif`.

Proyeksiya. Tayllar gradusda (uzunlik, kenglik) keladi. Ko‘rinish zonasi uchun **metr** kerak, shuning uchun 04-kundagidek, ularni Toshkent uchun metrli tizim bo‘lgan **UTM 42N zonasiga** (EPSG:32642) o‘tkazamiz.

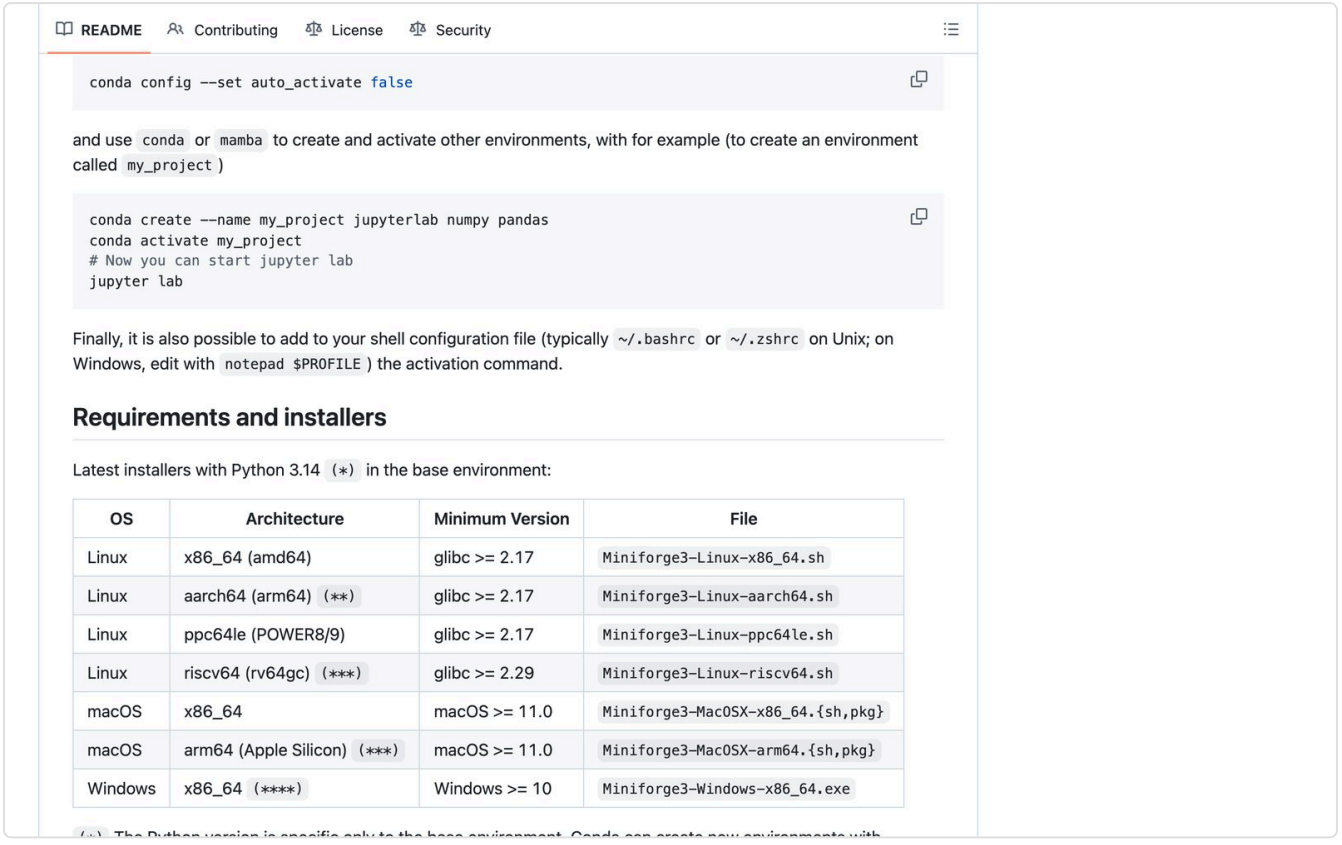
💾 1-qism · GDAL’ni o‘rnating va terminalni oching

🪟 Windows: u sizda allaqachon bor

Agar 04-kun uchun QGIS’ni o‘rnatgan bo‘lsangiz, GDAL u bilan birga keldi. **Start** menyusini oching, **OSGeo4W Shell** deb yozing va uni oching. Qora oyna paydo bo‘ladi: bu sizning terminalingiz.

🍎 macOS va 🐧 Linux: avval Miniforge, so‘ng GDAL’ni o‘rnating

Miniforge ilmiy vositalar uchun bepul o‘rnatuvchi. U GDAL’ni alohida "quti"ga joylaydi, shuning uchun kompyuteringizdagi hech narsani buzmaydi.



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```
conda config --set auto_activate false
```

and use `conda` or `mamba` to create and activate other environments, with for example (to create an environment called `my_project`)

```
conda create --name my_project jupyterlab numpy pandas
conda activate my_project
# Now you can start jupyter lab
jupyter lab
```

Finally, it is also possible to add to your shell configuration file (typically `~/.bashrc` or `~/.zshrc` on Unix; on Windows, edit with `notepad $PROFILE`) the activation command.

Requirements and installers

Latest installers with Python 3.14 `(*)` in the base environment:

OS	Architecture	Minimum Version	File
Linux	x86_64 (amd64)	glibc >= 2.17	`Miniforge3-Linux-x86_64.sh`
Linux	aarch64 (arm64) `(**)`	glibc >= 2.17	`Miniforge3-Linux-aarch64.sh`
Linux	ppc64le (POWER8/9)	glibc >= 2.17	`Miniforge3-Linux-ppc64le.sh`
Linux	riscv64 (rv64gc) `(***)`	glibc >= 2.29	`Miniforge3-Linux-riscv64.sh`
macOS	x86_64	macOS >= 11.0	`Miniforge3-MacOSX-x86_64.{sh,pkg}`
macOS	arm64 (Apple Silicon) `(***)`	macOS >= 11.0	`Miniforge3-MacOSX-arm64.{sh,pkg}`
Windows	x86_64 `(****)`	Windows >= 10	`Miniforge3-Windows-x86_64.exe`

1. **Terminal**ni oching (macOS: ⌘ **Space** ni bosing, *Terminal* deb yozing, Enter ni bosing).
2. Quyidagi ikki qatorni nusxalang, joylashtiring, har biridan keyin Enter ni bosing va barcha savollarga **yes** deb javob bering:

```
curl -L -O "https://github.com/conda-forge/miniforge/releases/latest/download/Minif
orge3-$(uname)-$(uname -m).sh"
bash Miniforge3-$(uname)-$(uname -m).sh
```

3. **Terminalni yoping va qayta oching.** Endi ichida GDAL bo‘lgan `gdal` nomli quti yarating va unga kiring:

   ```
   conda create -n gdal -c conda-forge gdal
   conda activate gdal
   ```

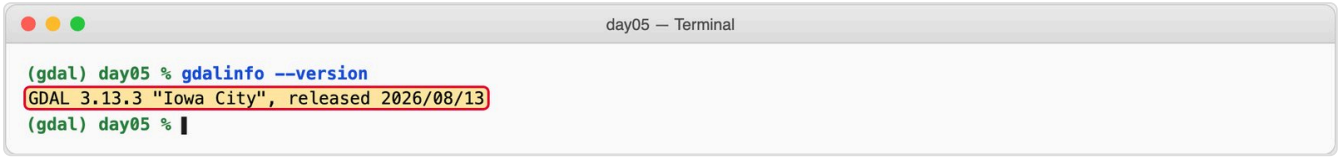
   Endi buyruq satringiz `(gdal)` bilan boshlanadi. Bu dars uchun har safar yangi Terminal ochganingizda, avval `conda activate gdal` deb yozing.

> 💡 Mac’da QGIS bilan birga ham ba’zi GDAL vositalari keladi, lekin bizning sinov Mac’imizda undagi `gdal_viewshed` ishlamadi. Miniforge sizga GDAL’ning eng yangi versiyasini beradi va har qanday kompyuterda bir xil ishlaydi. Windows’da ham Miniforge’dan foydalanishingiz mumkin (OSGeo4W Shell o‘rniga **Miniforge Prompt** ni oching).

✅ Tekshirib ko‘ring

Buni yozing va Enter ni bosing:

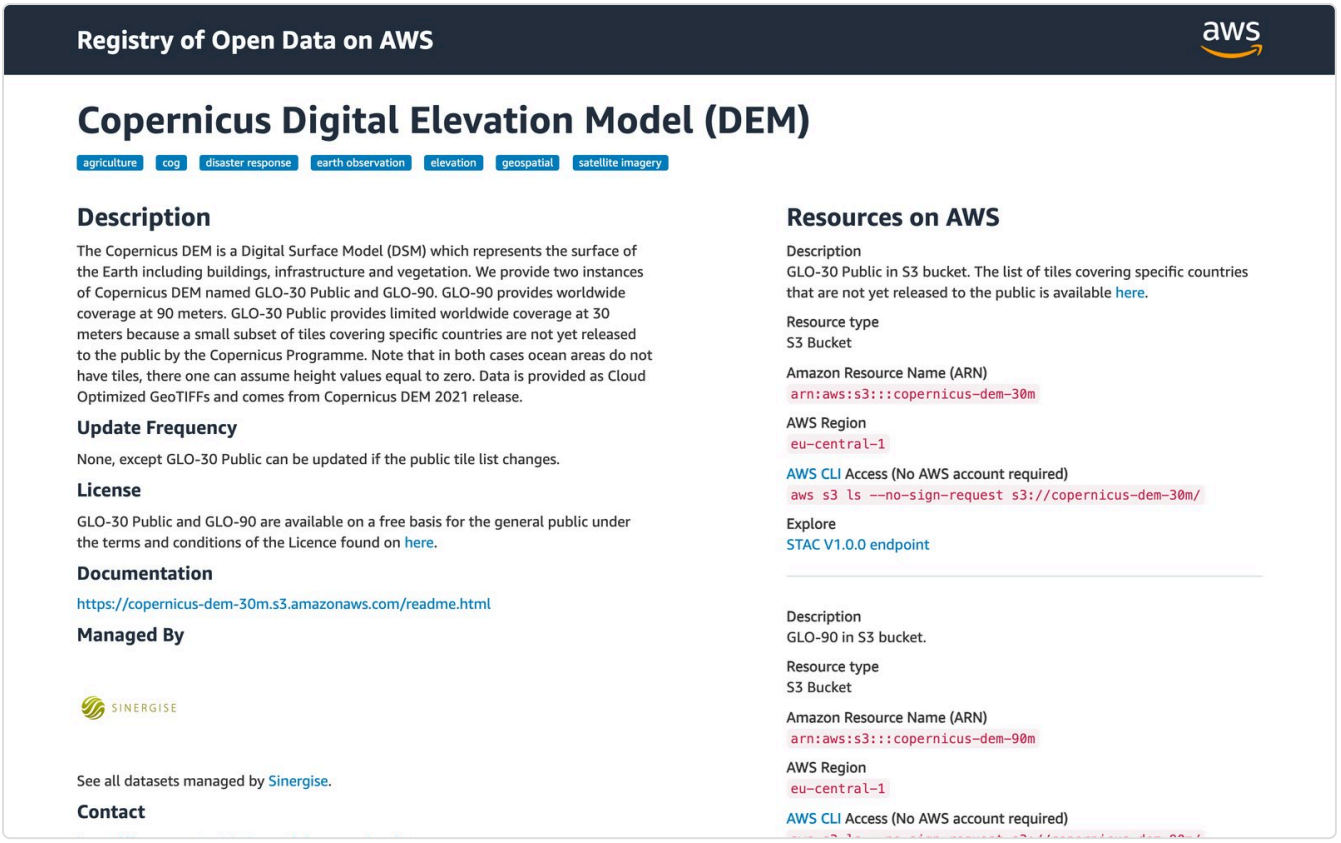
```
gdalinfo --version
```



Agar `GDAL 3.…` ni ko‘rsangiz, hammasi tayyor. Agar *command not found* chiqsa, Nimadir ishlamadimi? bo‘limiga qarang.

📥 2-qism · Sun’iy yo‘ldosh balandlik ma’lumotlarini yuklab oling

Copernicus DEM bepul va Amazon’ning ochiq ma’lumotlar xizmatida joylashgan, **akkaunt kerak emas**. Dunyo 1° × 1° o‘lchamli tayllarga bo‘lingan va har bir tayl o‘zining **pastki chap burchagi** bo‘yicha nomlanadi: `N41_00_E069` 41–42° shimoliy kenglik va 69–70° sharqiy uzunlikni qamraydi. Toshkent taxminan 41.3° N, 69.3° E da joylashgan.



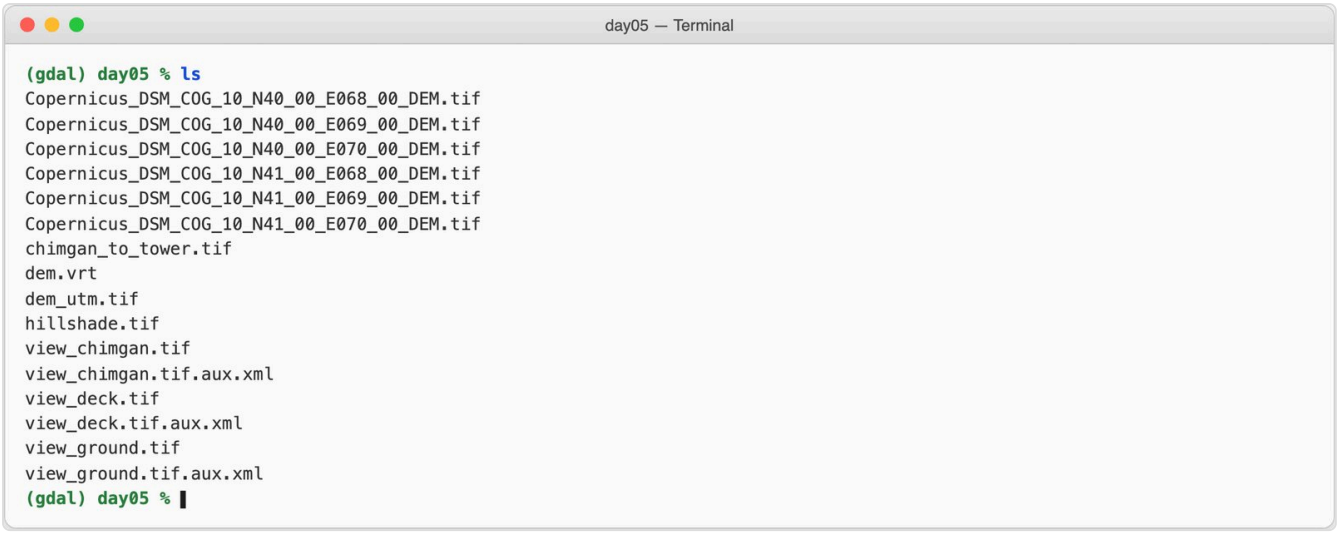
1. `day05` nomli papka yarating (masalan, *Downloads* ichida).
2. 6 ta taylni (har biri taxminan 40 MB) yuklab olish uchun har bir havolani bosing va ularni `day05` ga ko‘chiring:

Tayl	Qamrab oladi
N41 E068	Toshkentdan g‘arb, Chinoz
N41 E069	**Toshkent**, Chirchiq
N41 E070	**Chimyon**, Chorvoq suv ombori
N40 E068	Sirdaryo viloyati
N40 E069	Angren, Olmaliq, Bekobod
N40 E070	Qamchiq dovoni, Farg‘ona vodiysining shimoli

3. Terminalda `cd` ("change directory", ya’ni papkani almashtirish) yordamida papkaga kiring, so‘ng ichidagilarni ro‘yxat qilib ko‘ring (macOS/Linux’da `ls`, Windows’da `dir`):

```
cd Downloads/day05
ls
```

Windows’da: `cd %USERPROFILE%\Downloads\day05`, so‘ng `dir`.

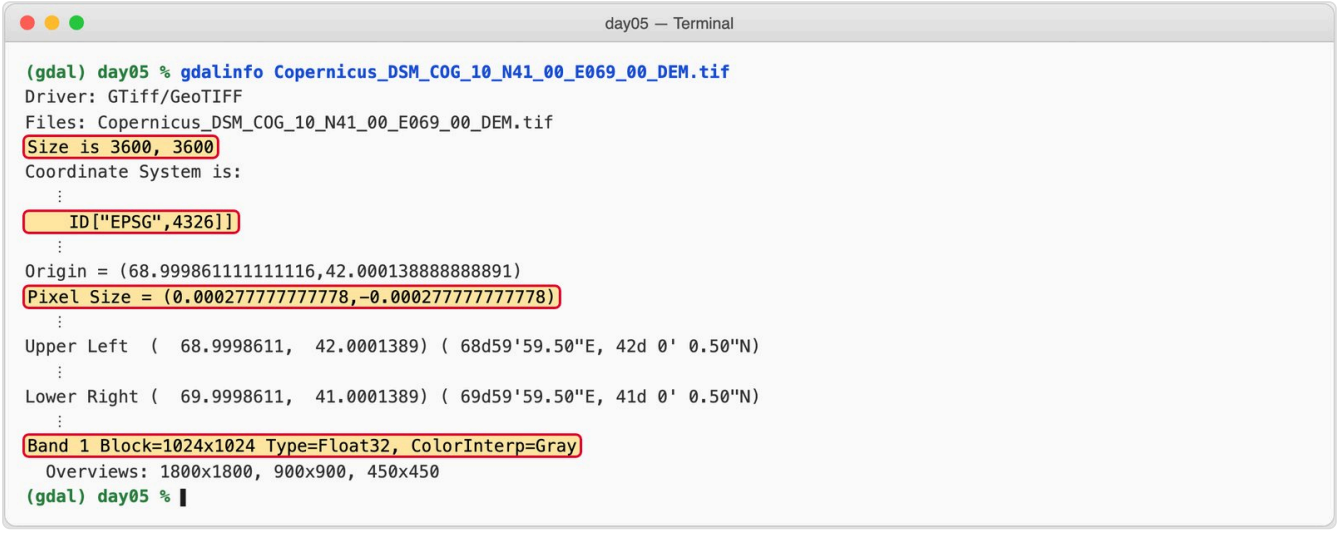


💡 **Boshqa shaharmi?** Tayllarni joyingiz joylashgan 1° kvadratning pastki chap burchagi bo‘yicha tanlang. Samarqand (39.65° N, 66.96° E) bu `N39_00_E066`, Buxoro `N39_00_E064`, Xiva `N41_00_E060`. Havoladagi ikkala raqamni ham o‘zgartiring.

🔍 3-qism · Tayl ichiga nazar soling (`gdalinfo`)

`gdalinfo` raster faylni ochmasdan turib u haqida hamma narsani aytib beradi:

```
gdalinfo Copernicus_DSM_COG_10_N41_00_E069_00_DEM.tif
```



Belgilangan qatorlarni o‘qing:

- **Size is 3600, 3600**: tayl 3,600 × 3,600 pikseldan iborat.
- **EPSG 4326**: koordinatalar gradusda (uzunlik, kenglik).
- **Pixel Size = 0.000277…**: bitta piksel gradusning 1/3600 qismi, taxminan 30 m.
- **Type=Float32**: har bir piksel o‘nli kasr son, ya’ni metrdagi balandlik.

🧩 4-qism · Tayllarni ulang va metrga o‘ting (`gdalbuildvrt`, `gdalwarp`)

Avval oltita taylni bitta **virtual** mozaikaga ulang. `*` belgisi "nomi `Copernicus_` bilan boshlanadigan har bir fayl" degani. Bu bir soniya oladi, chunki `.vrt` tayllarga ishora qiluvchi kichik matnli fayl, xolos:

```
gdalbuildvrt dem.vrt Copernicus_*.tif
```

So‘ng eng foydali GDAL vositasi `gdalwarp` yordamida hududimizni qirqib oling va uni metrga o‘tkazing:

```
gdalwarp -te 68.5 40.8 71.0 42.0 -te_srs EPSG:4326 -t_srs EPSG:32642 -tr 30 30 -r biline
ar dem.vrt dem_utm.tif
```

Parametr	Ma’nosi
`-te 68.5 40.8 71.0 42.0`	shu to‘rtburchak bo‘yicha qirqish: g‘arb, janub, sharq, shimol
`-te_srs EPSG:4326`	…va to‘rtburchak gradusda yozilgan
`-t_srs EPSG:32642`	UTM 42N zonasiga (metr) o‘tkazish
`-tr 30 30`	har bir pikselni aniq 30 × 30 m qilish
`-r bilinear`	qayta namunalashda balandliklarni silliqlash
`dem.vrt dem_utm.tif`	kirish, so‘ng chiqish fayli



Taxminan 15 soniyadan keyin sizda `dem_utm.tif` paydo bo‘ladi: 7,030 × 4,505 piksel, Sirdaryodan Chorvoq ortidagi tog‘largacha.

📍 5-qism · Pikseldan savol so‘rang (`gdallocationinfo`)

Teleminora ostidagi yer (41.3456° N, 69.2850° E) qanchalik baland? Katta Chimyon cho‘qqisi-chi (41.4952° N, 70.0583° E)? `-wgs84` parametri uzunlik va kenglikni yozish imkonini beradi:

```
gdallocationinfo -wgs84 dem_utm.tif 69.2850 41.3456
gdallocationinfo -wgs84 dem_utm.tif 70.0583 41.4952
```



Minora **477 m** balandlikda turibdi, Katta Chimyon cho‘qqisi esa **3,294 m** (rasmiy balandligi 3,309 m: 30 metrli piksel o‘rtacha qiymat bo‘lgani uchun o‘tkir cho‘qqilar biroz pastroq chiqadi).

> ⚠️ Uzunlik **birinchi** yoziladi (avval x, so‘ng y). Agar ularni almashtirib qo‘ysangiz, Hind okeanidagi biror joy haqida so‘ragan bo‘lasiz.

🔢 6-qism · Bitta nuqtani gradusdan metrga o‘tkazish (`gdaltransform`)

`gdal_viewshed` ga kuzatuvchining joylashuvi rasterdagi kabi metrda kerak. `gdaltransform` uni o‘tkazib beradi. `echo` ikkita sonni vositaga yuboradi:

```
echo 69.2850 41.3456 | gdaltransform -s_srs EPSG:4326 -t_srs EPSG:32642
echo 70.0583 41.4952 | gdaltransform -s_srs EPSG:4326 -t_srs EPSG:32642
```



Ularni (yaxlitlab) yozib qo‘ying:

Joy	x (sharqqa metr)	y (shimolga metr)
Toshkent teleminorasi	**523843**	**4577162**
Katta Chimyon	**588336**	**4594272**

⛰️ 7-qism · Soyali relyef (`gdaldem hillshade`)

Oddiy kulrang rangda balandliklarni ko‘rish qiyin. **Soyali relyef (hillshade)** yer yuzasini xuddi quyosh bir necha tomondan nur sochayotgandek yoritadi, shunda tog‘ tizmalari va vodiylar yaqqol ko‘zga tashlanadi:

```
gdaldem hillshade -multidirectional -z 2 dem_utm.tif hillshade.tif
```

`-multidirectional` bir necha tomondan tushadigan yorug‘likdan foydalanadi, `-z 2` esa tepaliklarni ikki baravar baland qilib ko‘rsatadi (tekis shaharda foydali).



👁️ 8-qism · Ko‘rinish zonasi (`gdal_viewshed`)

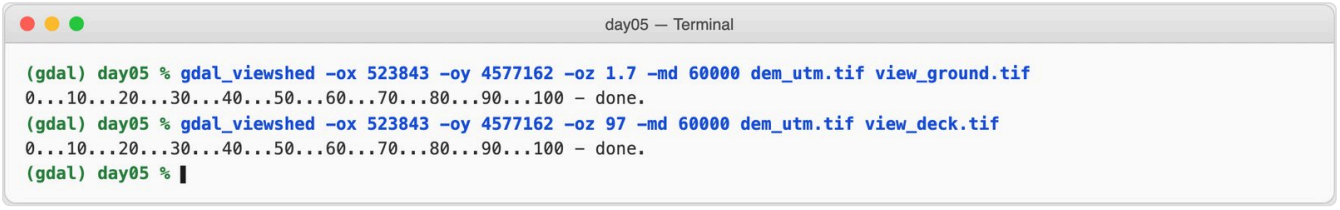
Endi asosiy savol. Avval minoraning **etagidan**, ko‘z balandligida:

```
gdal_viewshed -ox 523843 -oy 4577162 -oz 1.7 -md 60000 dem_utm.tif view_ground.tif
```

So‘ng 97 m balandlikdagi **kuzatuv maydonchasidan**:

```
gdal_viewshed -ox 523843 -oy 4577162 -oz 97 -md 60000 dem_utm.tif view_deck.tif
```

Parametr	Ma’nosi
`-ox 523843 -oy 4577162`	kuzatuvchi turgan joy (metrda, 6-qismdan)
`-oz 1.7` / `-oz 97`	ko‘zning yerdan balandligi, metrda
`-md 60000`	60 km gacha uzoqlikka qarash



Har biri bir soniyadan kam vaqt oladi. GDAL **Yerning egriligini** va havoda yorug‘likning egilishini hisobga olish uchun ko‘rish chizig‘ini biroz egadi, shuning uchun uzoqdagi joylar ham to‘g‘ri hisoblanadi.

📐 9-qism · Qancha joyni ko‘ra olasiz? (`gdalinfo -stats`)

Har bir ko‘rinish zonasi 4,001 × 4,001 pikselli kvadrat (har tomonga 60 km), unda ko‘rinadigan joylar **255**, yashirinlari **0**. GDAL’dan o‘rtacha qiymatni (**mean**) so‘rang:

```
gdalinfo -stats view_ground.tif
gdalinfo -stats view_deck.tif
```



O‘rtacha qiymatni kvadrat kilometrga aylantiring:

> **ko‘rinadigan maydon (km²) = o‘rtacha ÷ 255 × kenglik × balandlik × 900 ÷ 1,000,000**
> (900 m² bu bitta 30 × 30 m piksel.)

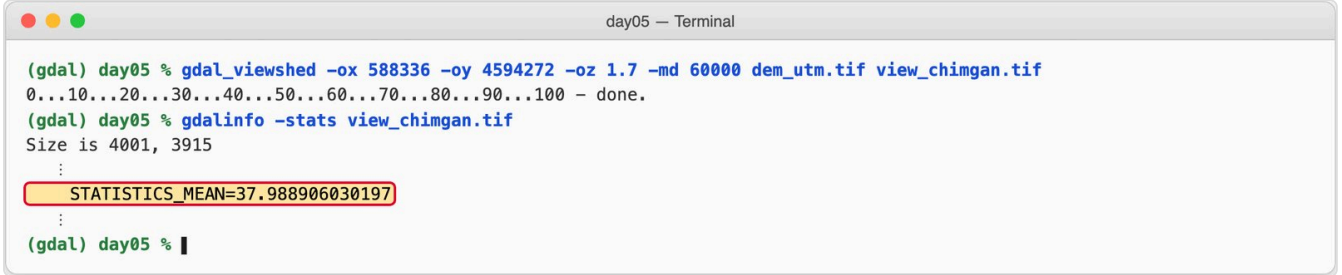
Kuzatuvchi	O‘rtacha	Kvadratdagi ulushi	Ko‘rinadigan maydon
Yer, 1.7 m	4.666	1.8 %	**264 km²**
Maydoncha, 97 m	29.416	11.5 %	**1,662 km²**

Yerdan turib qaraganda binolar va daraxtlar atrofingizdagi deyarli hamma narsani to‘sib qo‘yadi; faqat Toshkentga qaragan uzoqdagi tog‘ yonbag‘irlarini ko‘rasiz. Maydonchadan esa **olti baravar ko‘proq** joyni ko‘rasiz.

🏔️ 10-qism · Katta Chimyondan manzara: ular bir-birini ko‘ra oladimi?

Katta Chimyonga chiqing (kompyuterda):

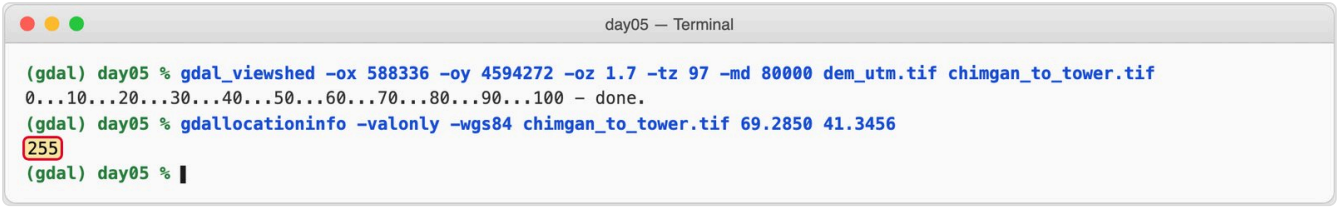
```
gdal_viewshed -ox 588336 -oy 4594272 -oz 1.7 -md 60000 dem_utm.tif view_chimgan.tif
gdalinfo -stats view_chimgan.tif
```



O‘rtacha qiymat 37.989, bu kvadrat esa 4,001 × 3,915 piksel (yuqori qismi ma’lumotlarimiz chegarasida qirqilgan), demak siz **2,100 km²** maydonni ko‘rasiz: Chorvoq suv ombori atrofidagi barcha yonbag‘irlar va Toshkent tomonga cho‘zilgan tekislikning keng yoyi (dumaloq chet bu bizning 60 km chegaramiz).

Cho‘qqidan teleminorani ko‘rish mumkinmi? `-tz` (nishon balandligi) parametri faqat yerni emas, balki *baland* narsani ko‘rish mumkinligini tekshiradi. 66.7 km uzoqlikdagi Chimyondan minoraning maydonchasini (97 m) ko‘rish mumkinmi, deb so‘rang, so‘ng minora joylashgan pikselni o‘qing:

```
gdal_viewshed -ox 588336 -oy 4594272 -oz 1.7 -tz 97 -md 80000 dem_utm.tif chimgan_to_tow
er.tif
gdallocationinfo -valonly -wgs84 chimgan_to_tower.tif 69.2850 41.3456
```



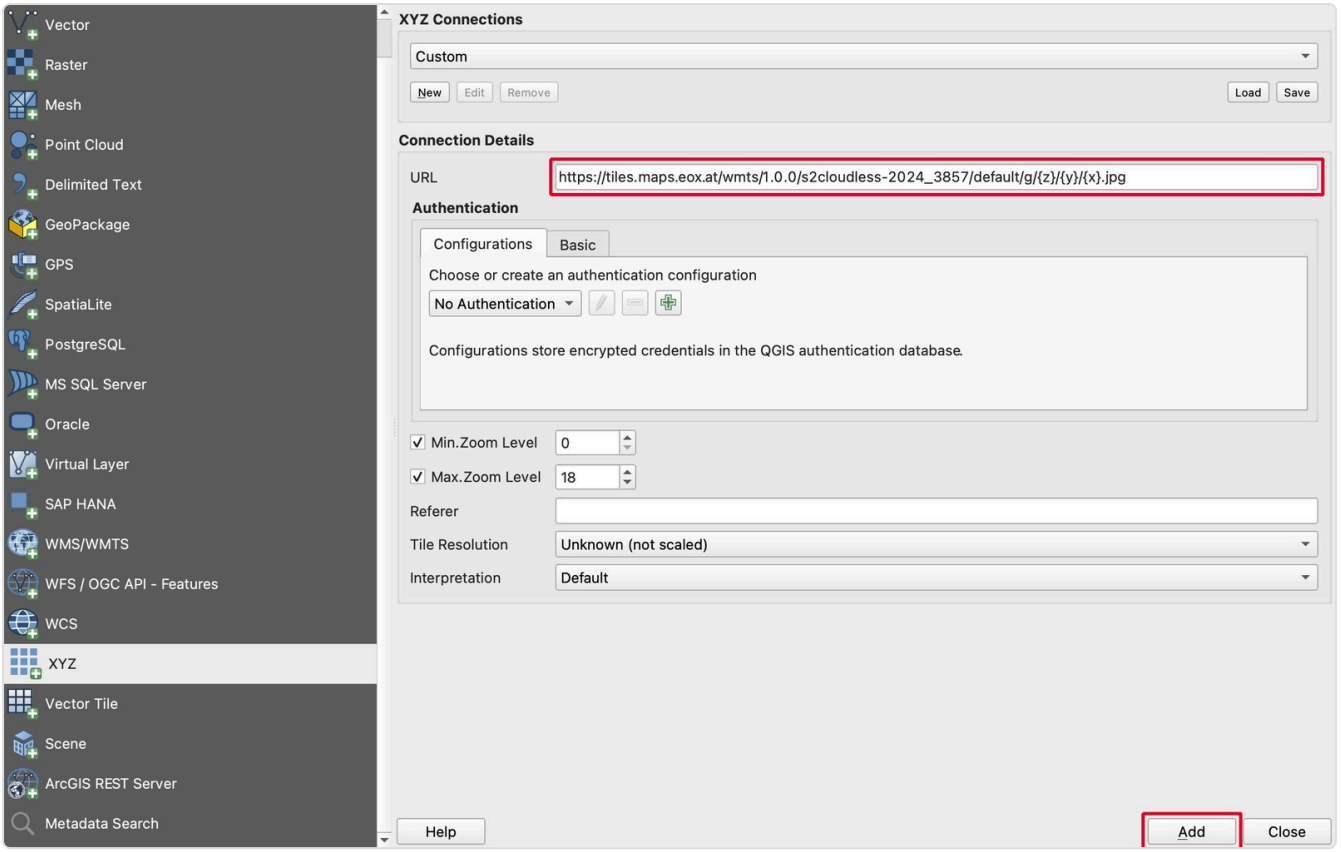
255 = ha! Havo ochiq kunda kuzatuv maydonchasidagi va Katta Chimyon cho‘qqisidagi odamlar bir-birini ko‘ra oladi.

🛰️ 11-qism · Natijani Sentinel-2 sun’iy yo‘ldosh tasviri ustiga qo‘ying (QGIS)

Endi chiroyli qilish vaqti. **QGIS** ni oching (04-kundan).

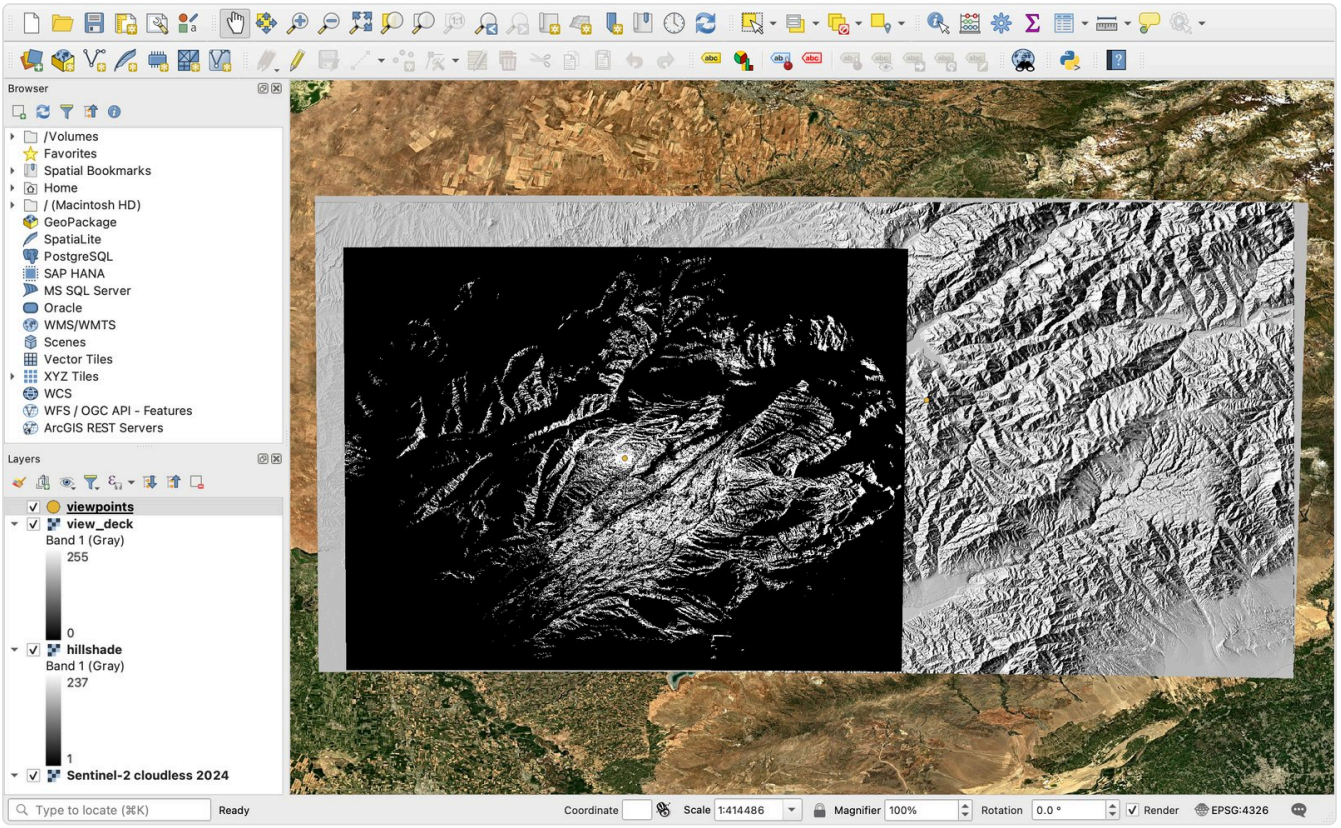
1. Butun hududning bulutsiz Sentinel-2 tasvirini qo‘shing. Layer → Add Layer → Add XYZ Layer… (Qatlam → Qatlam qo‘shish → XYZ qatlam qo‘shish) ga o‘ting. Quyidagi manzilni **URL** maydoniga joylashtiring va **Add** ni, so‘ng **Close** ni bosing:

```
https://tiles.maps.eox.at/wmts/1.0.0/s2cloudless-2024_3857/default/g/{z}/{y}/{x}.jpg
```

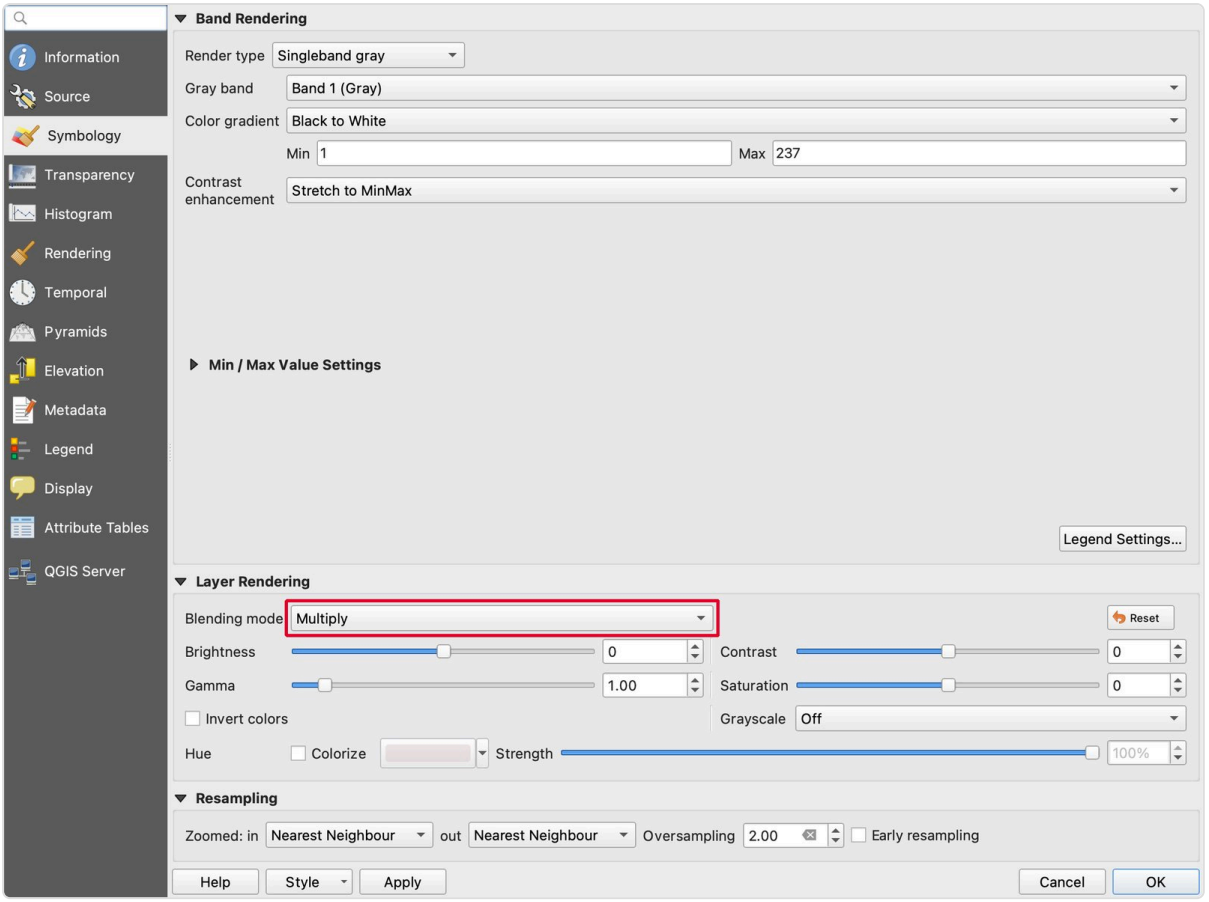


Bu **EOxCloudless 2024**: yil davomida olingan minglab Sentinel-2 suratlari, ulardan barcha bulutlar olib tashlangan.

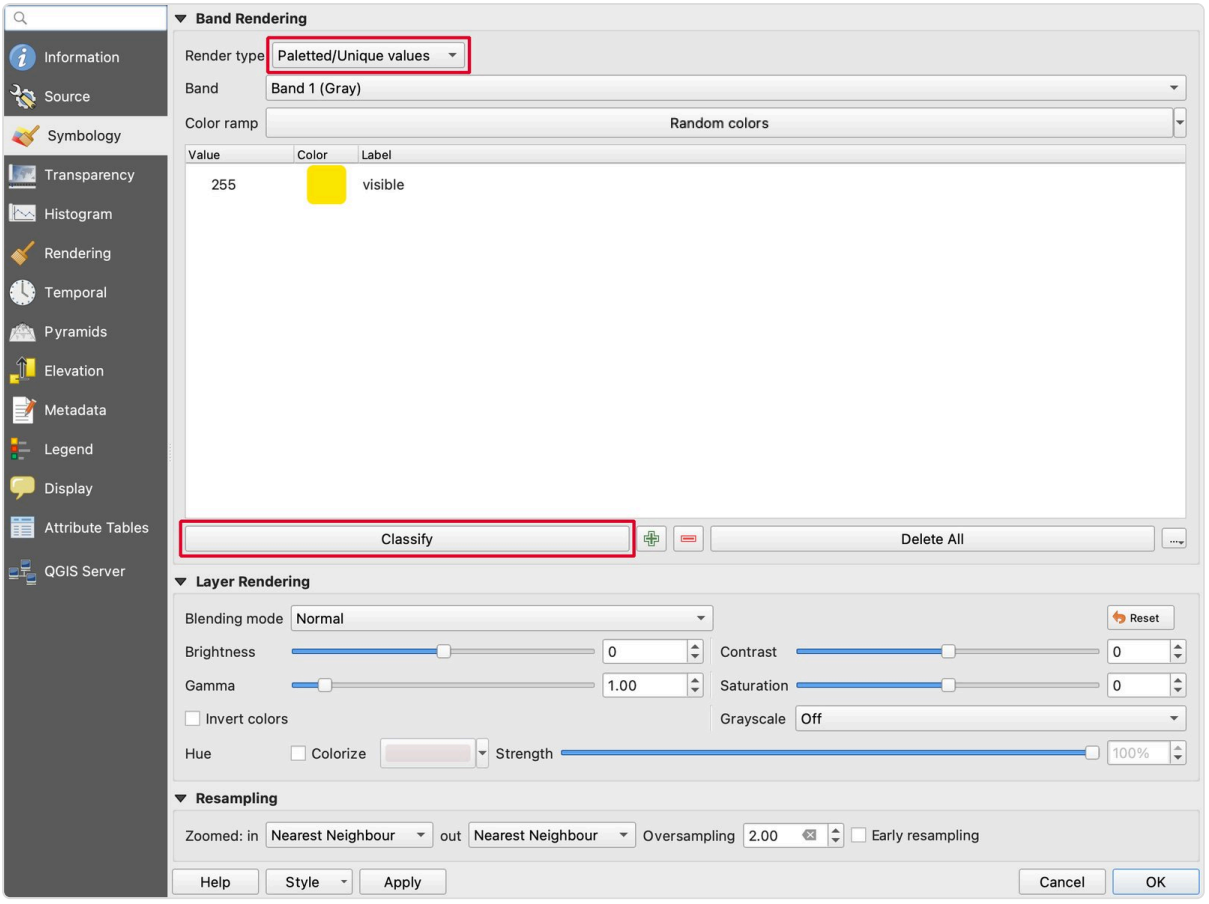
2. Fayllaringizni sudrab tashlang. `viewpoints.geojson` faylini (teleminora va Katta Chimyon ikki nuqta sifatida) `day05` ga yuklab oling. So‘ng `hillshade.tif`, `view_deck.tif` va `viewpoints.geojson` ni xaritaga sudrab tashlang. Ko‘rinish zonasi oq-qora rangda chiqadi: oq rang ko‘rinadigan joy.



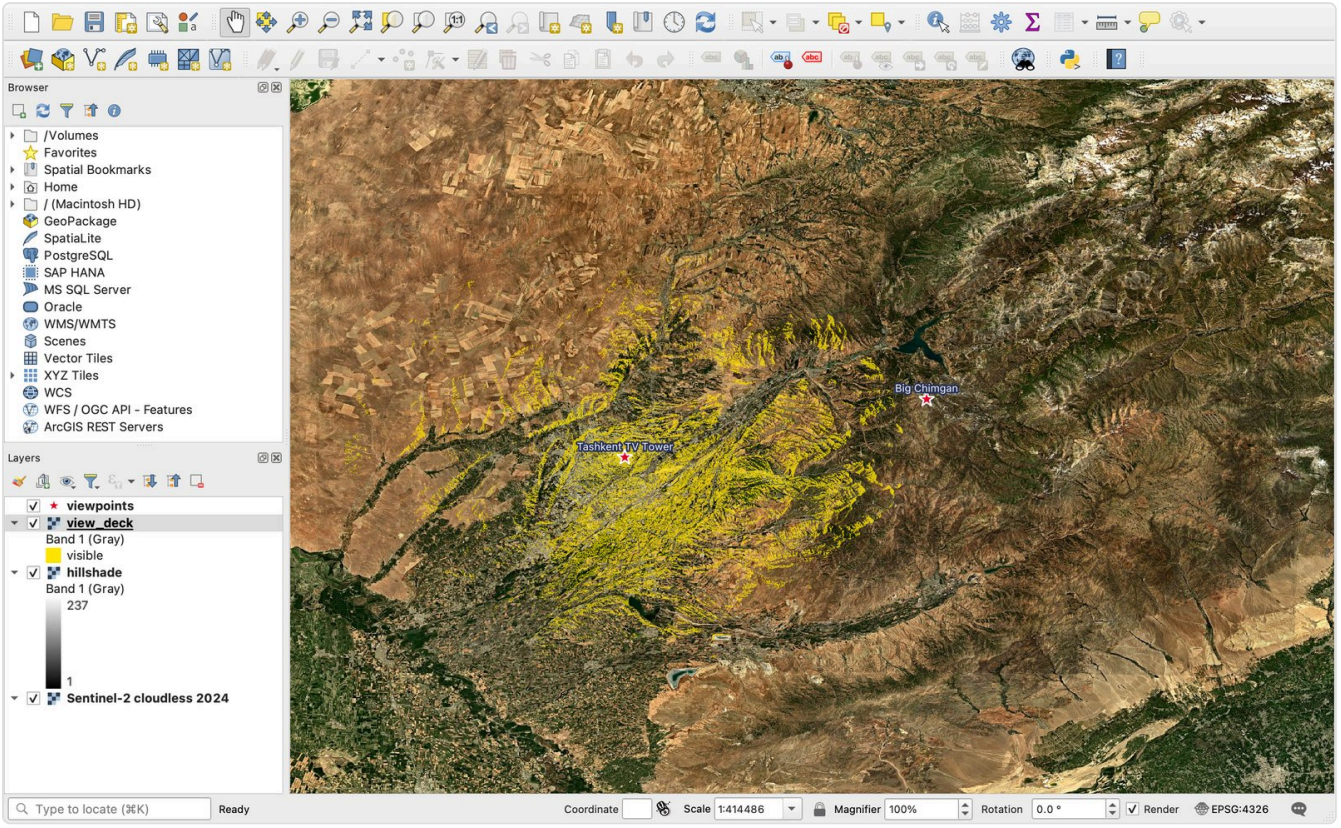
3. Relyefni suratga aralashtiring. Layers panelida **hillshade** ustiga ikki marta bosing → **Symbology** (Ko‘rinish uslubi). Pastda, *Layer Rendering* ostida **Blending mode** (aralashtirish rejimi) ni **Multiply** ga o‘rnating. So‘ng chapdagi **Transparency** (Shaffoflik) yorlig‘ini oching va **Global Opacity** ni taxminan 40 % qilib qo‘ying. **OK** ni bosing. Endi sun’iy yo‘ldosh tasvirida tog‘larning soyasi bor.



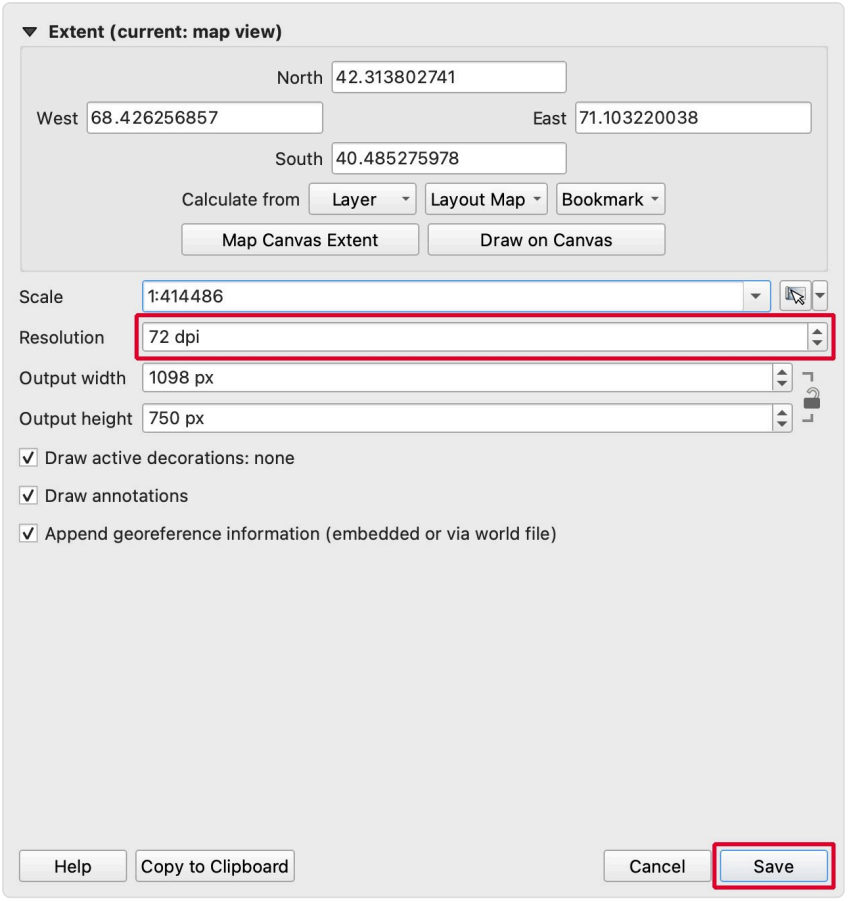
4. Faqat ko‘rinadigan joylarni ko‘rsating. view_deck ustiga ikki marta bosing → **Symbology**. **Render type** ni **Paletted/Unique values** (Palitrali/Noyob qiymatlar) ga o‘rnating va **Classify** (Tasniflash) ni bosing. Ikki qator paydo bo‘ladi: **0** va **255**. **0** qatorini tanlang va **minus** tugmasini bosing, shunda yashirin piksellar umuman chizilmaydi. **255** ning rangi ustiga ikki marta bosing va yorqin sariq rangni tanlang. **Transparency** yorlig‘ida **Global Opacity** ni 75 % qilib qo‘ying va **OK** ni bosing.



5. Kuzatuv nuqtalarini belgilang. viewpoints ustiga ikki marta bosing → **Symbology** → qizil **yulduzcha** (star) tanlang, o‘lchami 6. **Labels** (Yorliqlar) bo‘limida **Single Labels** ni **name** qiymati bilan tanlang va **Buffer** ostida bufer (matn atrofida oq yoki to‘q chegara) qo‘shing.



6. Rasmni saqlang. Project → Import/Export → Export Map to Image… (Xaritani rasm sifatida eksport qilish). **Resolution** ni **300 dpi** qilib qo‘ying va **Save** ni bosing.



Qolgan ikkita xaritani yasash uchun 4-qadamni `view_ground.tif` va `view_chimgan.tif` bilan takrorlang.

> 💡 Sarlavha, legenda va masshtab chizg‘ichi kerakmi? Xuddi 04-kun, 12-qismdagidek **Print Layout** dan foydalaning.

🌈 O‘zingizniki qiling: g‘oyalar

- **Minoralar kuzatuv minoralari bo‘lgan.** Buxorodagi **Kalon minorasi** (45.6 m, taxminan 39.7759° N, 64.4146° E, UTM 41N zonasi **EPSG:32641**, tayl `N39_00_E064`) yoki Xivadagi **Islom Xo‘ja minorasi** (57 m, taxminan 41.3774° N, 60.3600° E, EPSG:32641, tayl `N41_00_E060`) tepasidan ochiladigan manzarani hisoblang.
- **O‘z uyingiz.** QGIS’da binongizni toping, uning koordinatalarini o‘qing va `-oz` = qavatingiz × 3 m bilan ko‘rinish zonasini hisoblang. Derazangizdan Chimyon ko‘rinadimi? `-tz` dan foydalaning.
- **Minorani o‘stiring.** Teleminora ko‘rinish zonasini `-oz 1.7`, `25`, `50`, `97`, `200` bilan hisoblang va diagramma yasang: maydon va balandlik.
- **Minorani qayerlardan ko‘rish mumkin?** Savolni teskari qiling: kuzatuvchini hamma joyda yerga, nishonni esa tepaga qo‘ying. `gdal_viewshed -ox 523843 -oy 4577162 -oz 375 -tz 1.7 …` buyrug‘i odam (1.7 m) 375 metrli minoraning uchini ko‘ra oladigan barcha joylarni ko‘rsatadi: minora mo‘ljal sifatida qanchalik uzoqqacha xizmat qilishini shunday bilasiz.

- **Boshqa shaharlar, boshqa UTM zonalari:** Nukus EPSG:32640 · Buxoro, Navoiy, Urganch, Xiva EPSG:32641 · Toshkent, Samarqand, Namangan, Farg‘ona EPSG:32642 · Andijon EPSG:32643.

💾 Xaritangizni ulashing

Xaritangizni eksport qiling (11-qism), sarlavha qo‘shing va **#30DayMapChallenge** **#30DayMapChallengeUzbekistan** heshteglari bilan joylang. Iltimos, ma’lumotlar manbasini ko‘rsating (litsenziya talabiga ko‘ra matn inglizcha qoldirilgan):

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🆘 Nimadir ishlamadimi?

Muammo	Yechim
`command not found` / `is not recognized`	macOS/Linux: avval `conda activate gdal` deb yozing. Windows: oddiy Command Prompt emas, **OSGeo4W Shell** dan foydalaning.
`No such file or directory` / `does not exist`	Siz noto‘g‘ri papkadasiz. `ls` (yoki `dir`) deb yozing va tayllar shu yerdaligini tekshiring; `day05` ga o‘tish uchun `cd` dan foydalaning.
Windows: `gdalbuildvrt *` ni tushunmaydi	Uning o‘rniga oltita faylni nomma-nom sanab yozing: `gdalbuildvrt dem.vrt Copernicus_DSM_COG_10_N40_00_E068_00_DEM.tif Copernicus_DSM_COG_10_N40_00_E069_00_DEM.tif …`
`gdal_viewshed`: kuzatuvchi rasterdan tashqarida	Siz metr o‘rniga gradusdan foydalangansiz. `gdaltransform` dan olingan sonlarni ishlating (6-qism).
`Output dataset dem_utm.tif exists`	Eski faylni o‘chiring yoki `gdalwarp` buyrug‘iga `-overwrite` qo‘shing.
Ko‘rinish zonasi butunlay qora	Tartibni tekshiring: `-ox` bu x (523843), `-oy` bu y (4577162).
QGIS’da XYZ qatlam bo‘sh	Internet aloqangizni va URL’da aynan `{z}/{y}/{x}` borligini tekshiring.
Nusxalangan buyruqlar ishlamayapti	Har bir buyruqni **bitta qator** qilib joylashtiring. Ichiga "aqlli qo‘shtirnoq" (“ ”) kirib qolmaganiga ishonch hosil qiling.

📎 Mualliflik va manbalar

- **Balandlik:** Copernicus DEM GLO-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018, provided under COPERNICUS by the European Union and ESA; all rights reserved. Copernicus DEM litsenziyasi asosida bepul foydalanish mumkin.
- **Sun’iy yo‘ldosh tasviri:** EOxCloudless 2024, EOX IT Services GmbH (Contains modified Copernicus Sentinel data 2024), CC BY-NC-SA 4.0, notijorat maqsadlarda bepul.
- **Vositalar:** GDAL (MIT litsenziyasi), Miniforge, QGIS.
- Skrinshotlar: macOS terminalidagi GDAL 3.13, Mac’dagi QGIS 3.34. Balandliklar va maydonlar ushbu darsdagi buyruqlar bilan hisoblangan.